

TEXAS GULF COAST REGIONAL, TX, USA

WMO: 722527

Lat: 29.110N

Lon: 95.462W

Elev: 8

StdP: 101.23

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 12976

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.5	(c) 1.6	(d) -7.0	(e) 2.1	(f) 5.5	(g) -4.3	(h) 2.6	(i) 7.4	(j) 11.1	(k) 19.0	(l) 10.2	(m) 17.7	(n) 2.1	(o) 10	(p) 0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 9.7	(c) 35.0	(d) 25.5	(e) 33.9	(f) 25.5	(g) 33.0	(h) 25.5	(i) 27.3	(j) 31.4	(k) 26.9	(l) 31.0	(m) 26.6	(n) 30.7	(o) 4.3	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.8	28.6	26.0	21.4	28.6	25.5	20.7	28.5	85.9	31.4	84.0	31.0	82.6	30.4	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.6	8.6	7.9	-3.5	37.0	1.9	1.6	-4.8	38.2	-5.9	39.1	-7.0	40.1	-8.4	41.2
			-4.6	28.0	1.9	0.8	-5.9	28.6	-7.0	29.0	-8.1	29.5	-9.4	30.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.2	12.4	14.6	17.6	21.0	24.6	27.5	28.2	28.5	26.5	22.1	17.2	13.7
	DBStd	6.87	5.34	5.31	4.42	3.51	2.71	1.47	1.35	1.42	2.30	3.96	5.00	5.30
	HDD10.0	86	36	18	3	0	0	0	0	0	0	0	6	23
	HDD18.3	667	196	126	69	17	2	0	0	0	0	14	81	163
	CDD10.0	4168	109	146	238	329	454	525	564	574	496	374	221	137
	CDD18.3	1707	11	21	45	96	197	275	306	316	246	130	46	19
	CDH23.3	16411	29	54	152	548	1712	3001	3440	3611	2366	1147	275	75
	CDH26.7	6445	1	2	10	71	478	1264	1564	1717	968	332	36	2
Wind	WSAvg	3.4	3.7	4.0	4.0	4.2	4.0	3.2	2.6	2.4	2.5	2.9	3.2	3.6
Precipitation	PrecAvg	1251	103	72	81	96	100	118	97	101	143	120	109	100
	PrecMax	1794	293	218	223	230	263	303	323	303	375	353	366	246
	PrecMin	579	1	9	0	9	0	15	0	17	28	0	23	19
	PrecStd	303	70	53	54	64	70	68	78	80	93	93	80	65
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	26.5	27.8	30.0	32.5	35.5	35.7	36.8	36.2	32.3	29.2	26.9
		MCWB	20.0	20.5	20.7	21.7	24.0	24.6	25.5	25.5	24.7	24.4	23.0	21.8
	2%	DB	24.1	25.0	26.4	28.3	31.2	33.3	34.3	35.1	33.7	31.1	27.7	25.4
		MCWB	19.1	20.2	20.9	22.2	24.2	25.0	25.6	25.8	24.8	24.4	22.5	21.3
	5%	DB	22.7	23.6	25.0	27.4	30.2	32.5	33.4	34.0	32.5	29.9	26.3	23.9
		MCWB	19.2	19.7	20.3	22.1	24.0	25.1	25.7	25.9	24.9	23.6	21.6	20.8
	10%	DB	21.3	22.4	23.8	26.3	29.2	31.9	32.6	33.0	31.4	28.7	24.9	22.6
		MCWB	18.6	19.5	19.9	21.4	23.6	25.2	25.8	25.9	24.8	22.9	20.9	20.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.0	22.5	22.9	25.0	26.9	27.4	27.6	27.9	27.4	26.7	24.7	23.5
		MCDB	24.0	24.5	25.8	28.0	29.2	31.1	32.0	32.1	31.7	28.9	26.9	25.1
	2%	WB	21.2	21.6	22.2	23.9	25.8	26.8	27.1	27.3	26.8	25.8	23.9	22.5
		MCDB	23.0	23.5	24.6	26.6	28.6	30.4	31.4	31.7	30.7	28.6	26.2	24.3
	5%	WB	20.3	21.0	21.6	23.3	25.2	26.3	26.8	26.9	26.3	25.2	23.0	21.8
		MCDB	22.1	22.7	23.7	25.9	28.5	29.9	31.0	31.3	30.0	28.3	25.2	23.5
	10%	WB	19.2	20.2	21.1	22.8	24.6	25.8	26.4	26.5	25.8	24.5	22.1	20.6
		MCDB	21.0	21.9	23.0	25.2	27.9	29.7	30.6	31.0	29.5	27.5	24.3	22.3
Mean Daily Temperature Range	5% DB	MDBR	10.7	9.9	9.9	9.6	8.9	8.8	9.0	9.7	9.5	11.2	11.1	10.7
		MCDBR	10.8	9.6	9.2	9.3	9.4	9.8	10.2	10.8	10.8	10.7	10.4	10.9
		MCWBR	7.3	6.0	4.8	4.7	3.8	3.4	3.4	3.5	3.9	4.9	5.8	7.5
	5% WB	MCDBR	9.1	8.1	7.3	7.5	7.3	8.0	8.9	9.4	8.7	8.9	8.6	9.7
		MCWBR	7.4	6.0	4.5	4.3	3.4	3.5	3.5	3.5	3.7	4.9	6.0	7.6
Clear-Sky Solar Irradiance	taub	0.350	0.364	0.379	0.438	0.482	0.464	0.492	0.478	0.454	0.384	0.377	0.358	
	taud	2.497	2.468	2.412	2.248	2.158	2.267	2.207	2.248	2.283	2.469	2.435	2.480	
	Ebn at Noon	887	904	909	861	821	830	807	816	825	870	846	858	
	Edh at Noon	92	102	115	140	153	137	145	138	129	101	97	90	
All-Sky Solar Radiation	RadAvg	2.89	3.31	4.52	5.39	6.07	6.36	6.12	5.87	5.03	4.42	3.33	2.58	
	RadStd	0.29	0.55	0.40	0.42	0.41	0.46	0.58	0.44	0.52	0.48	0.34	0.31	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	+99

Nomenclature: See separate page